Qdislib — Circuit Cutting Cheat Sheet

Run circuits bigger than your device: cut, execute the pieces, recombine.

Qdislib is a distributed circuit-cutting library: it splits a quantum circuit into smaller subcircuits that fit your hardware, runs them independently (in parallel, even across backends or QPUs) and recombines the results classically. It is **backend-agnostic** — the same code drives Qiskit, Qibo, CUDA-Q, PennyLane and QiliSDK — and scales with PyCOMPSs.

WAYS TO CUT

Gate cutting replaces a two-qubit gate by a sum of single-qubit *measure-and-prepare* operations, splitting the circuit where that gate was. Cuts **any two-qubit gate** following its angle ($\gamma = 1 + 2|\sin\phi|$ per $rzz(\phi)$, 3 for a CZ). **Cost: 6^k** terms for *k* cuts.

Wire cutting cuts a qubit's wire (its timeline) into "measure here / re-prepare there". **Cost: 8^k** terms for *k* cuts.

Hadamard gate cutting (`gate_cutting_hadamard`) cuts a gate with one **local ancilla** per fragment instead of mid-circuit measurement — cuts **any** gate as one cut, and is the cheaper one for a gate with several interactions (iSWAP: γ 7 against 9).

Every cut multiplies the work **exponentially**, so use as few, and as well-placed, as possible. Preview the blow-up with `cutting_overhead(circ, cuts)` before you run.

THE WORKFLOW

Get one **expectation value** <O> in a single call, or use the **three-step** form to control how each piece runs (cluster, QPU):

```
cut = qd.find_cut(circ, gate_cut=True)
val = qd.gate_cutting(circ, cut, 'ZZZZ',
          shots=8192)      # None = exact

subs = qd.gate_cutting_subcircuits(
    circ, cut, 'ZZZZ')  # .circuits('qiskit')
vals = [qd.expectation_value(s,
    backend=None)  # or a QPU, per s
        for s in subs]
val = subs.reconstruct(vals)
```

`estimate(circ, 'ZZZZ', max_qubits=n)` does find-cut + cut + reconstruct in one call.

QUICKSTART

```
import Qdislib as qd
from qibo import Circuit, gates

circ = Circuit(4)
circ.add(gates.H(0))
circ.add(gates.CZ(0, 1))
circ.add(gates.CZ(1, 2))
circ.add(gates.CZ(2, 3))

cut = qd.find_cut(circ, gate_cut=True,
        wire_cut=False, seed=0)
val = qd.gate_cutting(circ, cut,
        observable='ZZZZ', shots=8192)
```

CORE API

`find_cut(circ, ...)`	Choose where to cut (returns it).
`gate_cutting(circ, cut)`	<O> via gate cutting.
`wire_cutting(circ, cut)`	<O> via wire cutting.
`gate_cutting_hadamard(...)`	<O> via the ancilla cut.
`circuit_cutting(circ, cuts)`	Any mix of gate/Hadamard/wire cuts.
`..._sampling(circ, cut)`	Full output distribution.
`..._subcircuits(...)`	3-step: pieces + .reconstruct().
`expectation_value(sub)`	Run one subcircuit.
`estimate(circ, ...)`	One-call cut + run + reconstruct.
`cutting_overhead(circ, cuts)`	Preview circuits and gamma.
`analytical_solution(...)`	Exact <O> (to check).
`draw_cut(circ, cuts, path)`	Fragments in colour, cuts marked.
`from_qasm(qasm, ...)`	Import any OpenQASM 2.

HADAMARD (ANCILLA) GATE CUT

An alternative gate cut (Harrow–Lowe, *arXiv:2403.01018*): each fragment carries **one extra ancilla** running a Hadamard test — **no mid-circuit measurement**, and it cuts **any** two-qubit gate (iSWAP, …) as a single cut.

```
val = qd.gate_cutting_hadamard(
        circ, ['ISWAP_3'], 'ZZ', shots=None)
# Monte-Carlo (bounded cost for many cuts):
val = qd.gate_cutting_hadamard(
        circ, cut, 'ZZ', num_samples=5000)
# three-step, runs on any backend / QPU:
subs = qd.gate_cutting_hadamard_subcircuits(
         circ, cut, 'ZZ')
val = subs.reconstruct(
  [qd.expectation_value(s) for s in subs])
```

Gates crossing one boundary share one ancilla per side by default (`joint=True`: $\gamma = 2^{k+1}-1$ instead of 3^k). Fragments are independent → they fan out as PyCOMPSs tasks like every other cut.

ANY MIX OF CUTS

Cut **gates** (measure-and-prepare or Hadamard) and **wires** on the **same circuit**, in any number. With `optimized=True` each fragment runs once per setting of its own cuts, so the cost **adds** over fragments instead of multiplying over cuts.

```
cuts = {'gate': ['CZ_2'], 'hadamard': [],
        'wire': [('RY_3', 'CZ_4')]}
val = qd.circuit_cutting(circ, cuts, 'ZZZ',
        optimized=True)
qd.cutting_overhead(circ, cuts, optimized=True)
```

SAMPLING THE DISTRIBUTION

Reconstruct the full **output distribution** `{bitstring: prob}` — to read out an image or inspect a state — instead of a single number:

```
dist = qd.gate_cutting_sampling(
         circ, cut, shots=200_000)
# three-step (parallel under PyCOMPSs):
subs = qd.gate_cutting_sampling_subcircuits(
         circ, cut)
dist = subs.reconstruct(
  [qd.sample_subcircuit(s) for s in subs])
```

`wire_cutting_sampling` too. Sampling carries a quasiprobability overhead, so it needs **more shots** than an expectation value.

FIND_CUT OPTIONS

`max_qubits=n`	Max qubits per subcircuit (device size).
`max_cuts=k`	Cap the number of cuts.
`gate_cut / wire_cut`	Which cut types to consider.
`coupling_map=`	Hardware-aware: a separating cut of the longest-routing gates (SparseCut, arXiv:2511.05492).
`mixed=True`	Gate, Hadamard and wire cuts priced by gamma (angle-aware, joint), within time_limit (10 s).
`seed=n`	Reproducible cut selection.

OBSERVABLES

Per-qubit Pauli string over **I X Y Z** (qubit-0-first): `'ZZZZ'`, `'ZIZIZ'`, `'XZZY'`. A Hamiltonian is a weighted sum:

```
[(0.5, 'ZZII'), (-1.0, 'IIZZ')]
# or  {'ZZII': 0.5, 'IIZZ': -1.0}
```

`I` ignores a qubit, also on a *cut* qubit.

BACKENDS

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`backend=`	Where it runs: 'qiskit', 'IBM_BACKEND', an object.
`subs.circuits(sdk)`	3-step: the subcircuits in any SDK.
`shots=None / seed=n`	Exact / reproducible, on every cut.
`build_backend(name, cfg)`	A configured backend (QPU, GPU Aer).
`available_backends()`	List registered backends.
`QdBackEnd(pool)`	Load-balance across devices.
`register_backend(...)`	Plug in your own (BaseBackend).

CACHING

```
cache = qd.default_circuit_cache(
          'memory', max_entries=5000)
qd.gate_cutting(circ, cut, cache=cache)
```

Keeps each fragment circuit's result, for every observable read from it (`cache_key='semantic'`: equivalent circuits too; `qd.use_cache(...)`: a session). Stores: `'memory'` / `'lmdb'` / `'redis'`. `set_log_level('DEBUG')` for diagnostics.

DISTRIBUTED WITH PYCOMPSS

Qdislib is built for HPC: every subcircuit is a PyCOMPSs `@task`, so the pieces run **in parallel across cores and nodes**. The three-step workflow fans them out automatically — the same code scales from a laptop to a supercomputer.

```
import pycompss.interactive as ipycompss
ipycompss.start()   # or:  runcompss app.py
subs = qd.gate_cutting_subcircuits(circ, cut)
vals = [qd.expectation_value(s) for s in subs]
val = subs.reconstruct(vals)   # auto-parallel
ipycompss.stop()
```

CHOOSING A METHOD

`gate_cutting`	Any 2-qubit gate; distribution too; no ancilla.
`gate_cutting_hadamard`	Any gate as one cut; joint groups; no mid-circuit measure, one ancilla per fragment.
`wire_cutting`	Cut a qubit's wire between two subcircuits.
`circuit_cutting`	Any mix of cuts on one circuit at once.
`*_sampling`	Reconstruct the full output distribution.

OVERHEAD AT A GLANCE

`gate cut / cut`	6 terms, $\gamma = 3$ (optimized: 5 circuits per side).
`wire cut / cut`	8 terms, $\gamma = 4$ (optimized: $2^n+1 + 4^n$ circuits).
`hadamard / cut`	γ(CZ)=3, k joint: $2^{k+1}-1$; $\to 1$ for weak gates.
`general gate`	up to 3 rzz rotations, each at its angle (one cut with the Hadamard method).

SCALE THE PARALLELISM

Every subcircuit is an independent task. Extra knobs for big fan-outs:

`sync=False`	Return futures; overlap several cuts (wait once).
`circuits_per_task=n`	Fewer, longer tasks for small circuits.
`optimized=True`	Gate / wire / mix: a sum over fragments, not a product over cuts.
`num_samples=n`	Every cut: draw n configurations, not all of them.

GOTCHAS

- Cost is exponential in #cuts (6^k / 8^k) — fewer, well-placed cuts win; `optimized=True` turns much of it into a sum; preview with `cutting_overhead`.
- Weak gates are cheap to cut. For a gate with several interactions prefer `gate_cutting_hadamard`: Qdislib warns when it is cheaper.
- Sampling needs **more shots** than an expectation value (quasiprobability overhead).
- `error=True` gives a standard error at no device cost; `total_shots=` spends one budget.
- Use `seed=` for a reproducible `find_cut`.

REAL HARDWARE

Route the subcircuits to GPUs or QPUs — the cutting code is unchanged:

```
ibm = qd.build_backend('IBM_BACKEND', cfg)
qd.gate_cutting(circ, cut, backend=ibm)   # IBM
qd.gate_cutting(circ, cut,
        backend='BSC_QPU_BACKEND')        # BSC QPU
```

`qd.pack`, or `pack=True` on the IBM backend, runs many subcircuits in one job.

BRING YOUR OWN CIRCUIT

Pass a **Qibo, Qiskit, CUDA-Q, PennyLane or QiliSDK** circuit — Qdislib converts it through a neutral DAG that speaks every backend (U3/CU3/… included). Or import OpenQASM 2:

```
circ = qd.from_qasm(open('c.qasm').read())
qd.gate_cutting(circ, cut, backend='qiskit')
```

LEARN MORE

Full docs, the thirteen notebooks and three applications and the API reference:

`Docs`	qdislib.readthedocs.io
`Source`	github.com/bsc-wdc/qdislib
`Notebooks`	notebooks/ — open in Colab or Binder.
`estimate(...)`	One call: find-cut + cut + reconstruct.

SANITY CHECK

Validate a cut against the exact value on a small instance before you scale it up (cutting reconstructs the same <O>, within shot noise):

```
exact = qd.analytical_solution(circ, 'ZZ')
val   = qd.gate_cutting(circ, cut, 'ZZ',
                        shots=8192)
assert abs(val - exact) < 0.05
qd.draw_cut(circ, cut, 'cut.png')   # see it
```